SQL - THE MASTER LIBRARIAN

INSERT, UPDATE, DELETE & JOINs

Planrs Academy | Grade 6 Session 2

Managing & Connecting Your Digital World

Previously in SQL...

In our last session, we learned that a database is like a digital library made of tables. We learned:

- **Rows** are records (like a student).
- **Columns** are fields (like a name or age).
- **Primary Keys** are unique IDs (like Aadhaar numbers).

> But how do we add a new student? What if someone changes their address? Today, we learn the commands to control our data!

The Zomato Mystery

How does a delivery partner know your name, your address, and exactly which Biryani you ordered at the same time?

> The information isn't in one giant, messy list. It's stored in different tables! Zomato uses **SQL commands** to connect them instantly.

Today, you will learn the same commands used by the world's biggest apps!

The "New Admission" (INSERT)

When a new student joins Planrs Academy, we don't buy a new register. We just **INSERT** a new row!

```
INSERT INTO Students (RollNo, Name, Class)
VALUES (105, 'Arjun', '6B');
```

This command adds Arjun's details into the 'Students' table perfectly.

The "Address Change" (UPDATE)

What if a student moves to a new house? We don't want to delete them and start over. We just **UPDATE** their specific info.

```
UPDATE Students
SET Address = 'Bandra, Mumbai'
WHERE RollNo = 105;
```

Crucial Rule: Always use the **WHERE** clause! If you forget it, the database might change everyone's address to Mumbai!

The "Library Exit" (DELETE)

When a book is too old or gets lost, we remove it from our digital records using **DELETE**.

```
DELETE FROM Books
WHERE BookID = 'B-501';
```

Just like with UPDATE, we use the **WHERE** clause to make sure we only remove the specific book we intended to.

Meet the JOIN

A **JOIN** is like a bridge between two tables. It lets us see a complete picture of our data.

> Imagine one table has **Student Names** and another has **Exam Scores**. They both share a **Roll Number**.
>
> The JOIN uses that Roll Number to connect the name to the score!

Connecting Students & Scores

Table 1: Students

RollNo	Name
1	Priya

Table 2: Scores

RollNo	Marks
1	95

Result after JOIN:

Name	Marks
Priya	95

Why not just use one table?

Why do we keep data in different tables and then JOIN them?

- **Neatness:** Tables stay small and easy to manage.
- **Speed:** Computers can find things faster in small tables.
- **No Mistakes:** We only have to change a phone number in one place (the Students table) instead of ten different lists.

UPI Payments & SQL

When you use UPI to pay for a snack, a massive SQL operation happens!

1. The system **JOINs** your Phone Number to your Bank Account.
2. It **UPDATEs** your balance (subtracts the money).
3. It **INSERTs** a new record in your Transaction History.

Database Manager Challenge!

Which command would you use for these school scenarios?

1. A new student, Diya, joins the school. __________
2. A student changes their phone number. __________
3. A student graduates and leaves the school. __________
4. You want to see who borrowed 'The Jungle Book'. __________

Cricket Stats Manager

Let's update our Cricket Team from the last session!

Scenario: Rohit just scored 100 runs! How do we tell the database?

```
UPDATE CricketTeam
SET Runs = Runs + 100
WHERE PlayerName = 'Rohit';
```

Now the team's records are perfectly accurate!

What We Mastered Today

- **INSERT:** For adding new friends to the list.
- **UPDATE:** For fixing details like phone numbers.
- **DELETE:** For removing things we no longer need.
- **JOIN:** The bridge that connects our tables together.

You are now a true **Master Librarian** of data!

What's Next?

The apps you use every day—YouTube, Instagram, and Minecraft—all use these four commands to work. Every time you like a post, an **INSERT** or **UPDATE** is happening!

Keep exploring, keep building, and stay curious!

THANK YOU!

Master Librarian: Level Complete!

See you in the next session!